Statistical Learning and Data Analysis Module B

Shrinkage and Regularization

Roberta Siciliano

Statistics, tèchnè-loghìa, analysis of data

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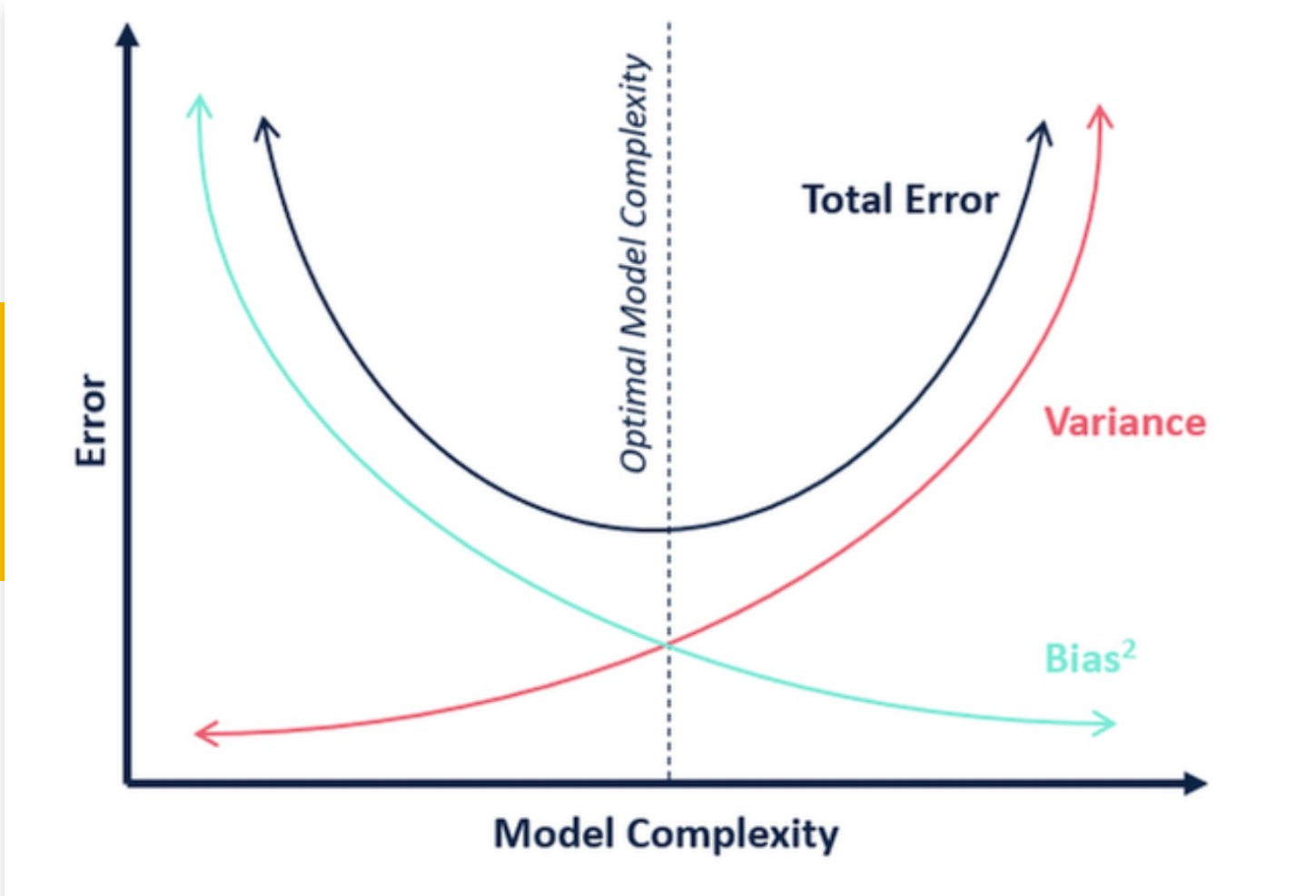

A thought

$$E\left[y_0 - \hat{f}(x_0)\right]^2 = var\left(\hat{f}(X_0)\right) + \left[bias\left(\hat{f}(x_0)\right)\right]^2 + var(\varepsilon)$$

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Why to use alternative fitting procedures

- **Prediction Accuracy**
  - When $n \gg p$ OLS tend to have low variance performing well on test observations.
  - If n is not so much larger than p OLS have high variance yielding to overfitting.
  - When $p > n$ there is not unique estimate, the variance is infinite.
  - ✓ By *constraining* or *shrinking* the estimated coefficients, we can reduce the variance at the cost of a negligible increase in bias.
- **Model Interpretability**
  - Some predictors can be irrelevant in multiple linear regression yielding to unnecessary complexity in the resulting model.
  - ✓ By automatically performing *feature/variable selection*, we can simplify the model specification ensuring a low MSE.

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Shrinkage methods

- As an alternative to the model selection techniques, we can fit a model containing all p predictors using tools that ***regularizes*** the coefficient estimates, or equivalently, that ***shrinks*** the coefficient estimates towards zero.
- We are talking about RIDGE regression and LASSO regression.

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Why would *shrunk* coefficients be better?

- The idea is to perform a linear regression, while *regularizing* or *shrinking* the coefficients estimates $\hat{\beta}$ toward 0.
- Why?
  - This introduces bias but may significantly decrease the variance of the estimates. If the latter effect is larger, this would *decrease the test error*.
  - There are Bayesian motivations to do this: *the prior tends to shrink the parameters.*

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Ordinary Least Squares recall

- OLS regression solves the following optimization:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- The minimum loss is equal to $RSS = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \sum_{j=1}^{p} \hat{\beta}_j x_{ij} \right)^2$.

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Ridge Regression

- Ridge regression solves the following optimization:

$$\hat{\beta}^R = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}$$

- where $\lambda \geq 0$ is a ***tuning parameter***, to be determined separately, and the term $\lambda \sum_{j=1}^{p} \beta_j^2$ is called **shrinkage penalty**.
- The tuning parameter modulates the importance of fit *versus* shrinkage.
- We find an estimate $\hat{\beta}^R$ for many values of λ and then choose it by cross-validation.

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fx Scaling the variables

- The linear regression model:

$$E(Y|X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j + \cdots + \beta_p X_p$$

- Scaling the variables has no effect on the OLS fit of the model:
  - multiplying a predictor X_j by a constant value c can be compensated by dividing the regression coefficient $\hat{\beta}_j$ by c, i.e., after doing this we have the same RSS.
- In ridge regression, this is not true.
- What do we do?
  - ➢ Scale each variable such that it has sample variance 1 before running the regression.
  - ➢ This prevents penalizing some coefficients more than others.

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The role of the tuning parameter

- The shrinkage penalty $\lambda \sum_{j=1}^{p} \beta_j^2$ is small when $\beta_1, \dots, \beta_p$ are close to zero, and so it has the effect of shrinking the estimates of β_j towards zero.
- The tuning parameter λ serves to control the relative impact of these two terms on the regression coefficient estimates.
- $\sqrt{\sum_{j=1}^{p} \hat{\beta}_j^2} = \|\hat{\beta}\|_2$ measures the distance of $\hat{\beta}$ from zero.
- $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$ is a measure to compare ridge regression estimates $\hat{\beta}_\lambda^R$ and OLS estimates $\hat{\beta}$.
- If $\lambda = 0$, then $\hat{\beta}_\lambda^R = \hat{\beta}$, so their ℓ_2 norms are the same (i.e., $\|\hat{\beta}_\lambda^R\| / \|\hat{\beta}\|_2 = 1$)
- If $\lambda \to \infty$, then $\hat{\beta}_\lambda^R$ will approach zero.
- Note that the $\hat{\beta}_\lambda^R = 0$ only if $\lambda = \infty$ (i.e., $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2 = 0$).

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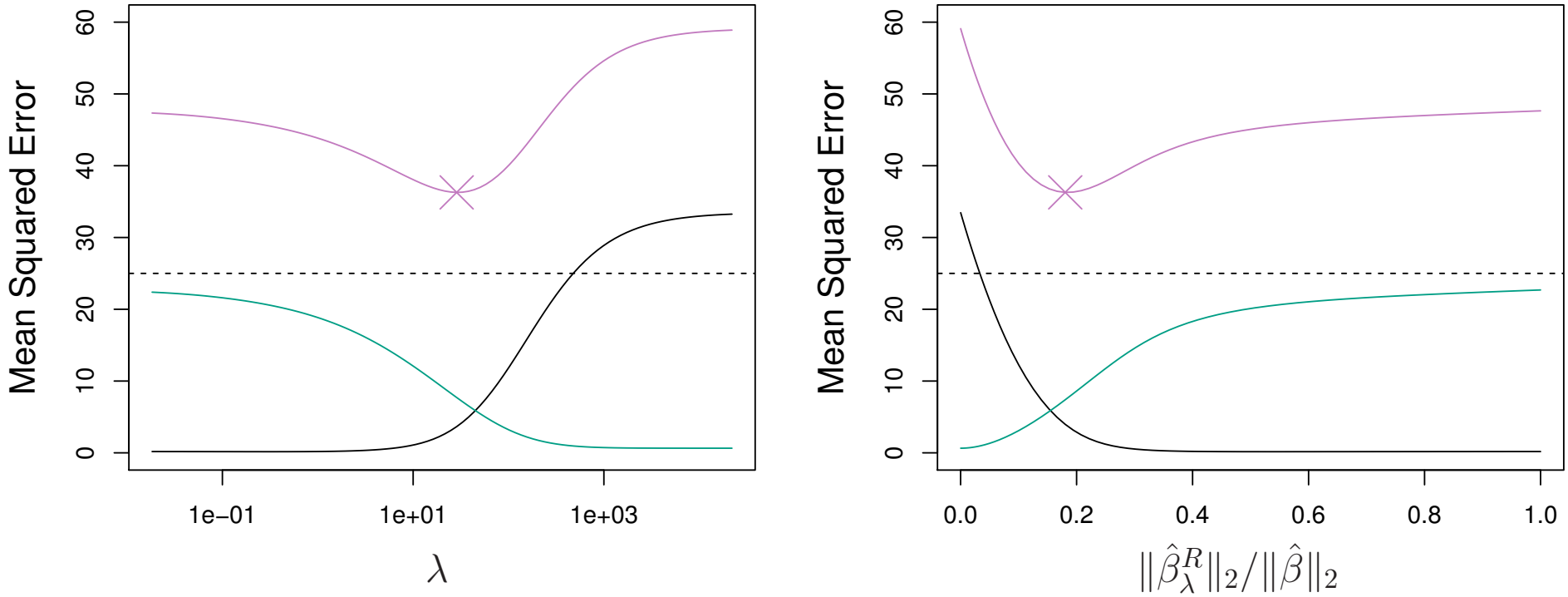
Bias-Variance Trade-off

Simulated data set: Squared bias, **variance**, and **test MSE** for the **ridge regression predictions**. *The horizontal dashed line indicate the minimum MSE (irreducible error).*



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Ridge Regression

- Using centered predictors, the estimation of the intercept is the sample mean of the response variable. The remaining coefficient get estimated by a ridge regression without intercept using centered predictors. Suppose that the input matrix is $\widetilde{\mathbf{X}}$ having p rather than $(p+1)$ columns, we can write the loss criterion in matrix form as:

$$RSS(\lambda) = \left(\mathbf{y} - \widetilde{\mathbf{X}}\beta\right)'\left(\mathbf{y} - \widetilde{\mathbf{X}}\beta\right) + \lambda\beta'\beta$$

- The Ridge regression solutions are easily seen to be

$$\hat{\beta}^R = \left(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}} + \lambda\mathbf{I}\right)^{-1}\widetilde{\mathbf{X}}'\mathbf{y}$$

  where $\mathbf{I}$ is the identity matrix.

- The solution is again a linear function of y, adds a positive constant to the diagonal $\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}}$ before inversion. This makes the problem nonsingular even if $\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}}$ is not of full rank, solving the problem of collinearity (Hoerl and Kennard, 1970).

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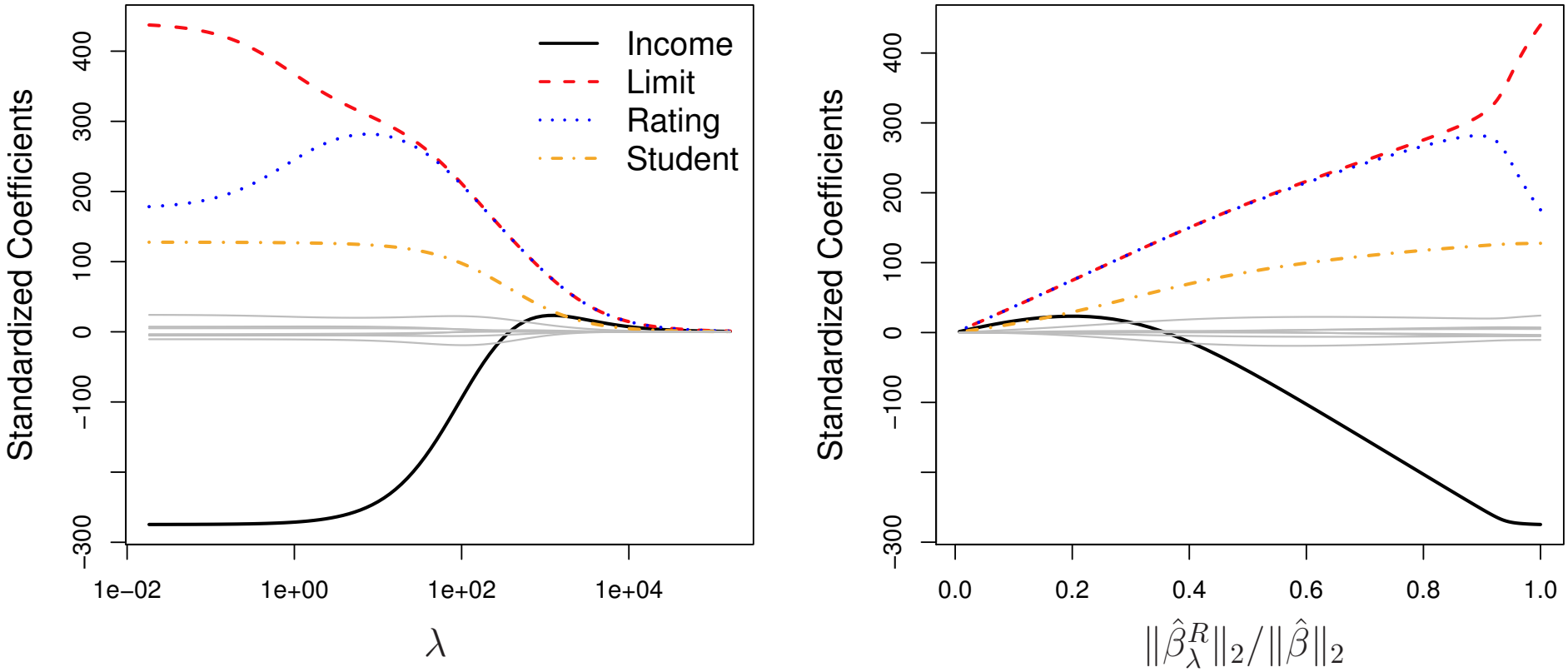
Credit data set: The standardized ridge regression coefficient estimates



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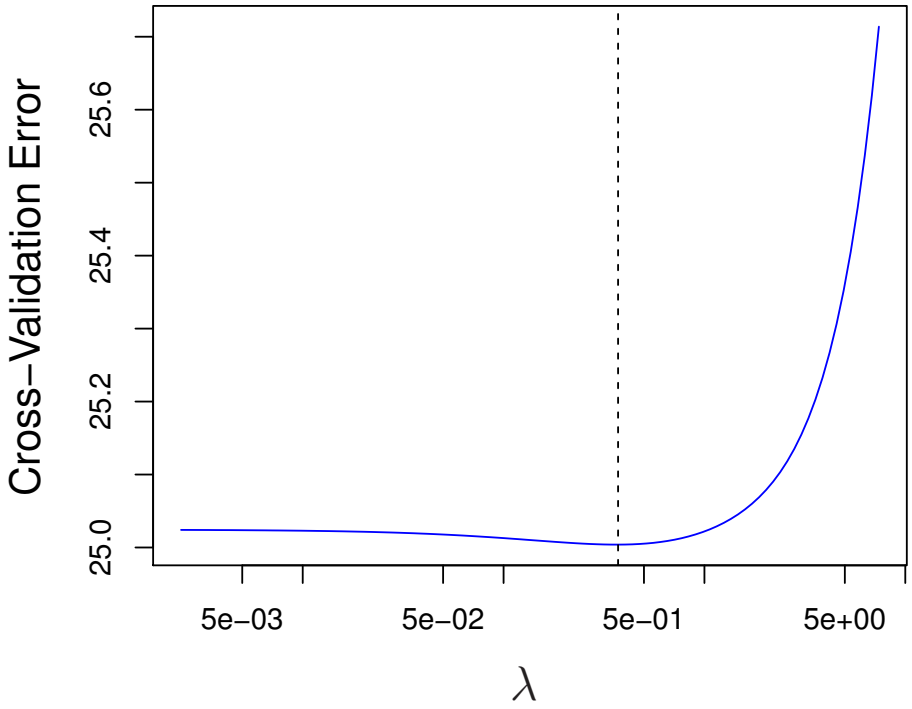
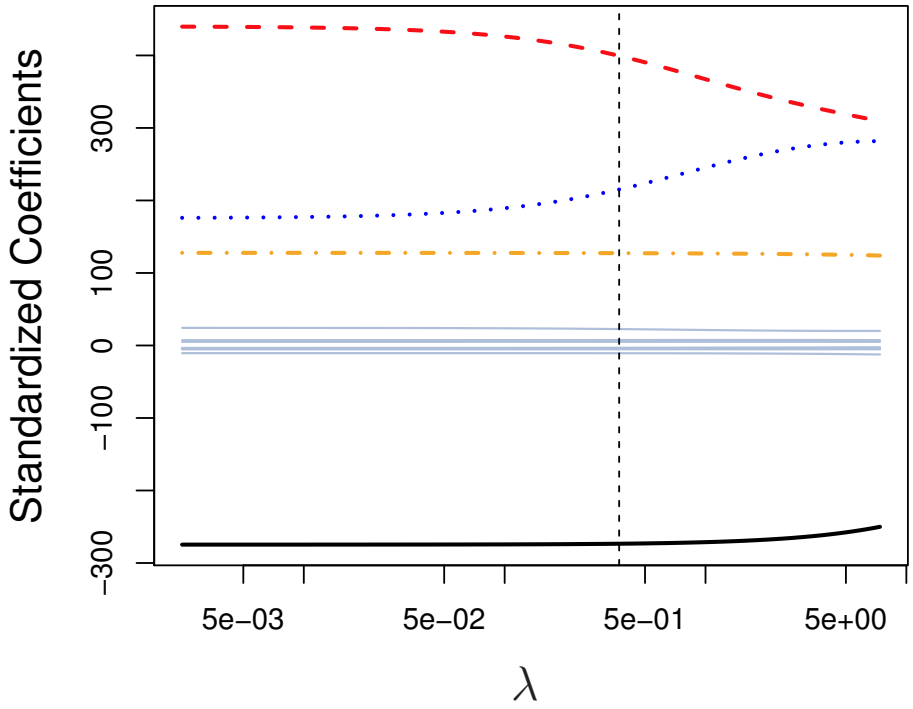
Credit data set: Predictors: **Income**, **Limit** **Rating,** Student

Cross-validation errors and Ridge Regression Coefficients with various λ.

Dash line indicate the value of selected λ





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LASSO Regression (Tibshirani, 1996)

- The **LASSO (Least Absolute Shrinkage and Selection Operator)** regression is an alternative to ridge regression. The objective function is

$$\hat{\beta}^L = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

- The LASSO regression, as the ridge regression, shrinks the coefficient estimates towards zero.
- Instead of the ℓ_2 norm we have the ℓ_1 norm of β or $\|\beta\|_1$.
- However, in the case of the LASSO, the penalty has the effect of ***forcing some of the coefficient estimates to be exactly equal to zero*** when the tuning parameter λ is sufficiently large.
- Hence, the LASSO regression performs variable selection.

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LASSO Properties

- ✓ Extremely fast algorithms for solving lasso problems (with many loss functions). Pathwise coordinate descent via GLMNET (Friedman, Tibshirani, 2010) exploits sparsity, active-set convergence, strong rules, and more, to rapidly compute entire solution path on a grid of values of λ.
- ✓ LARS Package in R (Efron, Johnstone, Tibshirani, 2002).
- ✓ With large p provides convenient subset selection, taking *leaps* rather than single steps.
- ▪ Since coefficients are both selected and regularized, can suffer from shrinkage bias.

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Some remarks on LASSO Regression

- The Lasso is an attractive regularization method for high dimensional regression.
- It combines variable selection with an efficient computational procedure.
- However, the rate of convergence of the Lasso is slow for some sparse high dimensional data, where the number of predictor variables is growing fast with the number of observations.
- Moreover, many noise variables are selected if the estimator is chosen by cross-validation.

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Why using LASSO instead of Ridge?

Ridge regression shrinks all the coefficients to a non-zero value.

The LASSO shrinks some of the coefficients all the way to zero.

Alternative convex to best subset selection or stepwise selection!

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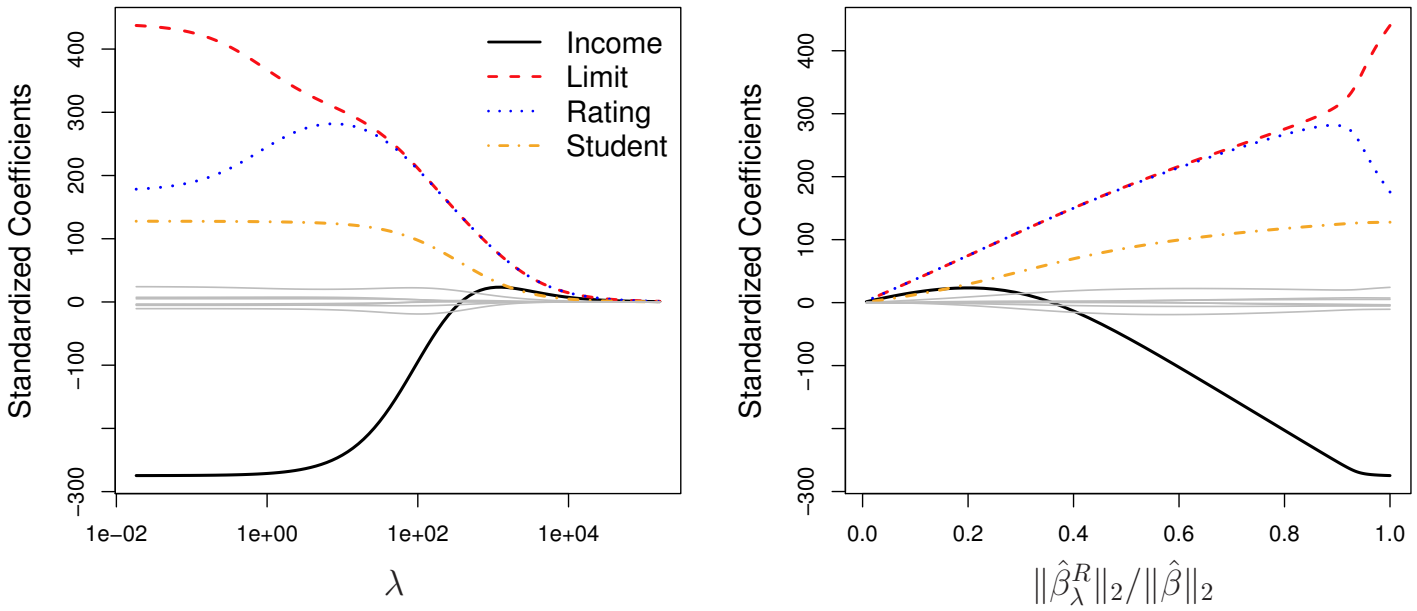
Credit data set: The standardized **ridge regression** coefficient estimates

A lot of pesky small coefficients throughout the regularization path.



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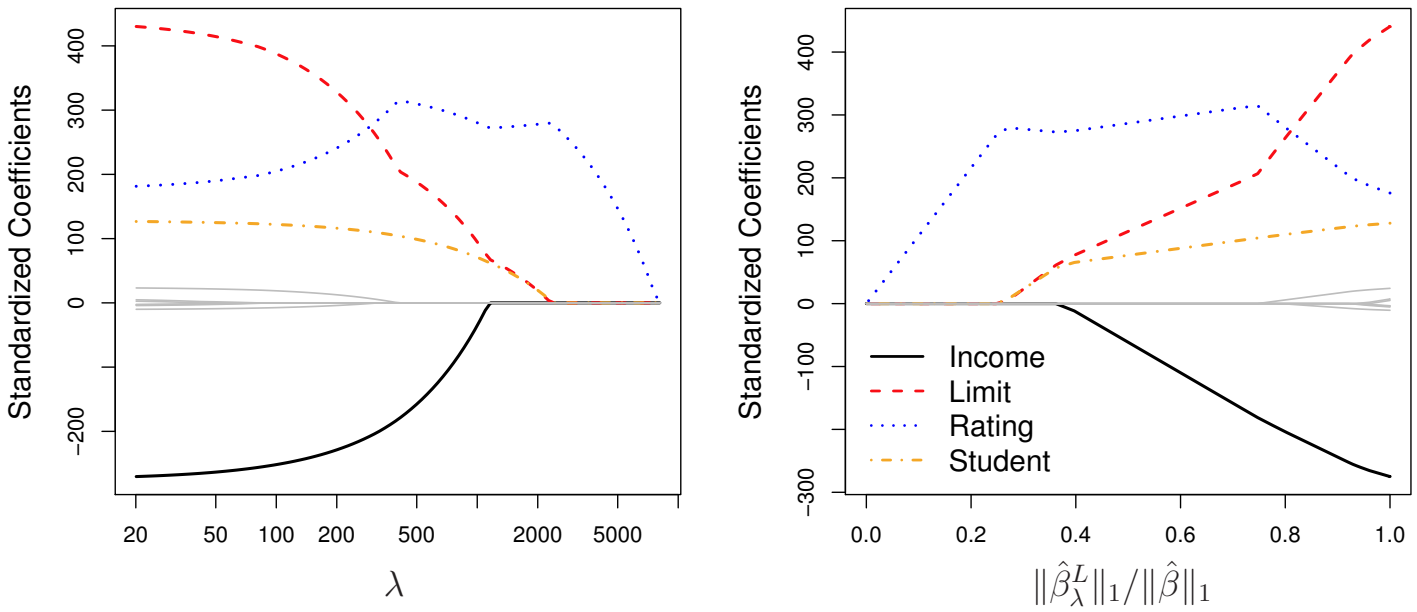
Credit data set: The standardized **LASSO regression** coefficient estimates

Those coefficients are shrunk to zero.



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Another formulation for Ridge and Lasso Regression

$$\hat{\beta}^R = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij}\right)^2 \text{subject to } \sum_{j=1}^{p} \beta_j^2 \leq s$$

$$\hat{\beta}^L = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij}\right)^2 \text{subject to } \sum_{j=1}^{p} |\beta_j| \leq s$$

- In LASSO we find the set of coefficients yielding to the minimum RSS subject to the constraint that there is a **budget $\boldsymbol{s}$** for how large $\sum_{j=1}^{p} |\beta_j|$ can be. If s is extremely large, then this budget is not very restrictive and so the coefficient estimates can be large, thus yielding to OLS.
- Similarly, it holds for ridge regression where $\sum_{j=1}^{p} \beta_j^2$ do not exceed the **budget $\boldsymbol{s}$**.

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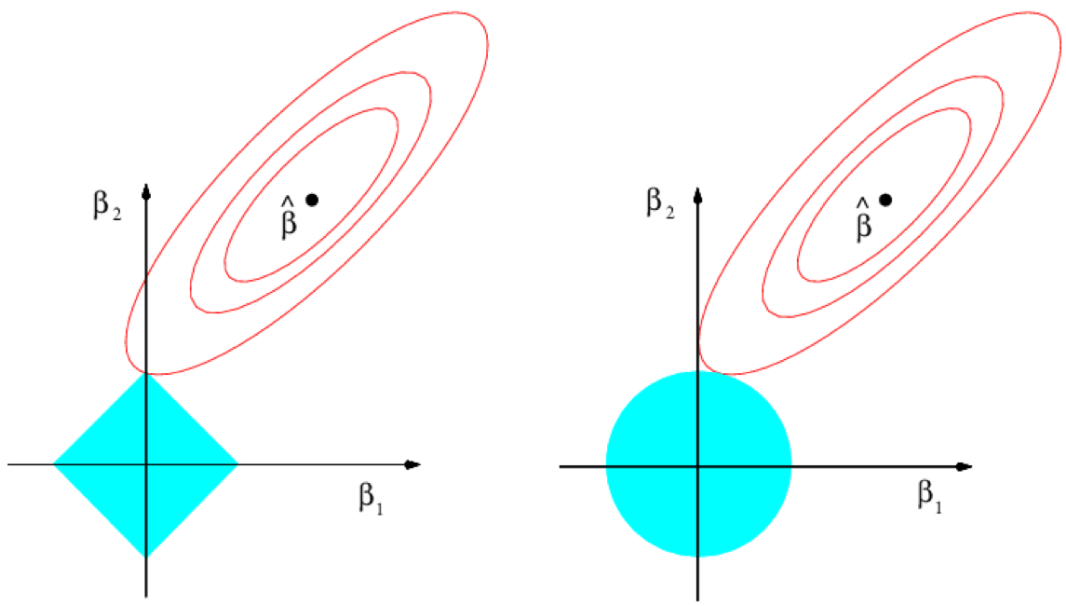
The variable selection property of LASSO ($p = 2$)



- **Ellipses** are the contours of the RSS.
- **Solid square area** are the constraint region for the lasso regression $|\beta_1| + |\beta_2| \leq s$
- **Solid circle area** are the constraint region for the ridge regression $\beta_1^2 + \beta_2^2 \leq s$

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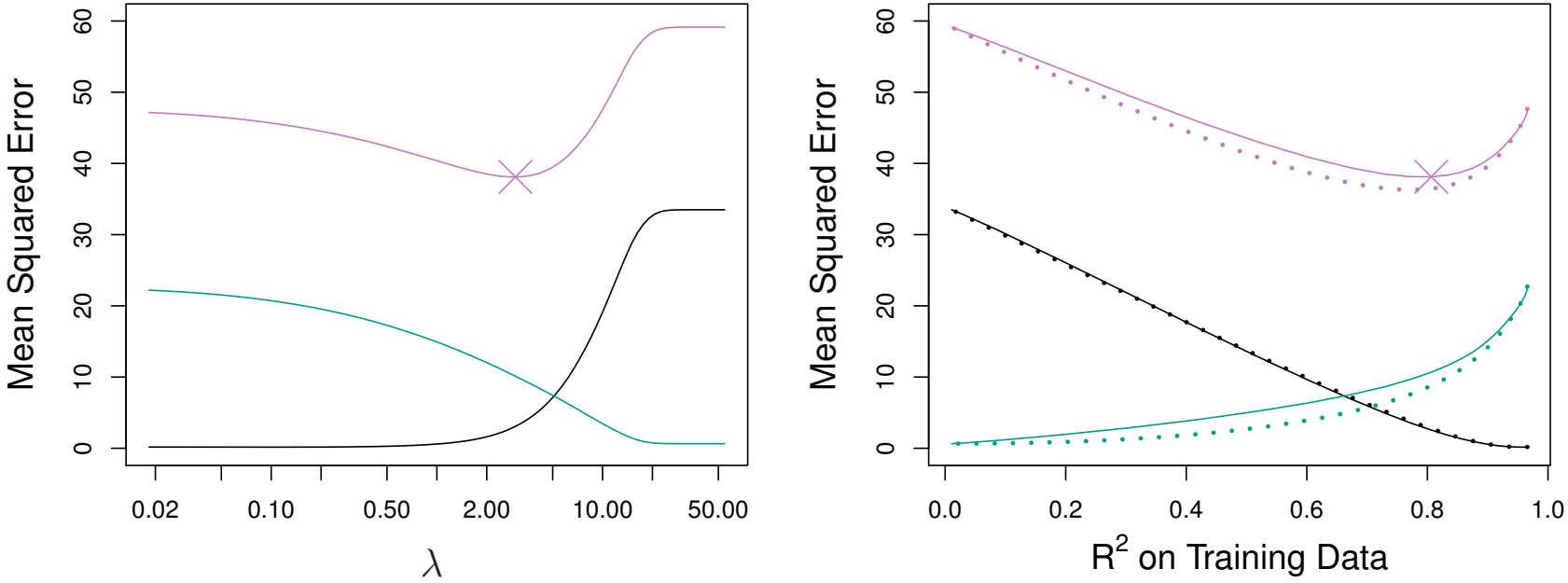
When is the LASSO better than Ridge?

Simulated data set: most of the coefficients of 45 predictors are non-zero



LASSO (solid lines) – Ridge (dotted lines)
The crosses in both plots indicate the lasso model for which the MSE is smallest.
Squared bias, **variance**, and **test MSE**

The bias is about the same for both methods.
The variance of Ridge is smaller, so is the MSE.

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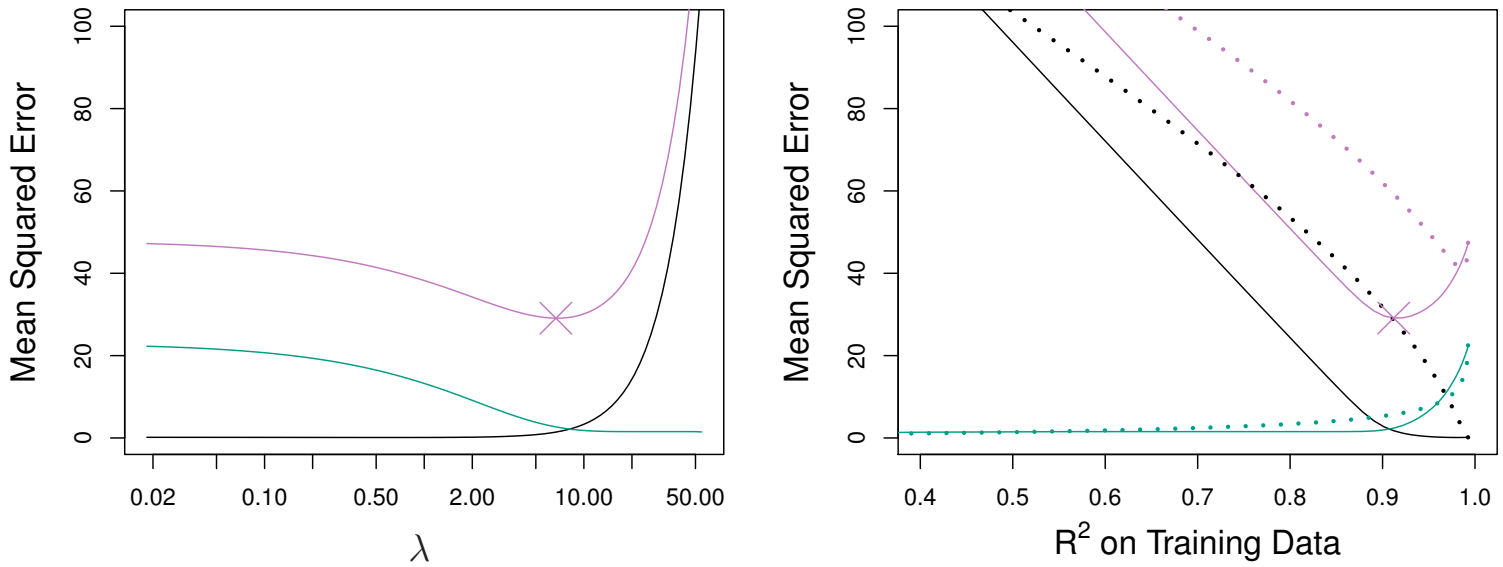
When is the LASSO better than Ridge?

Simulated data set: only 2 coefficients are non-zero



LASSO (solid lines) – Ridge (dotted lines)
The crosses in both plots indicate the lasso model for which the MSE is smallest.
Squared bias, **variance**, and **test MSE**

The bias, variance and MSE are lower for the LASSO.

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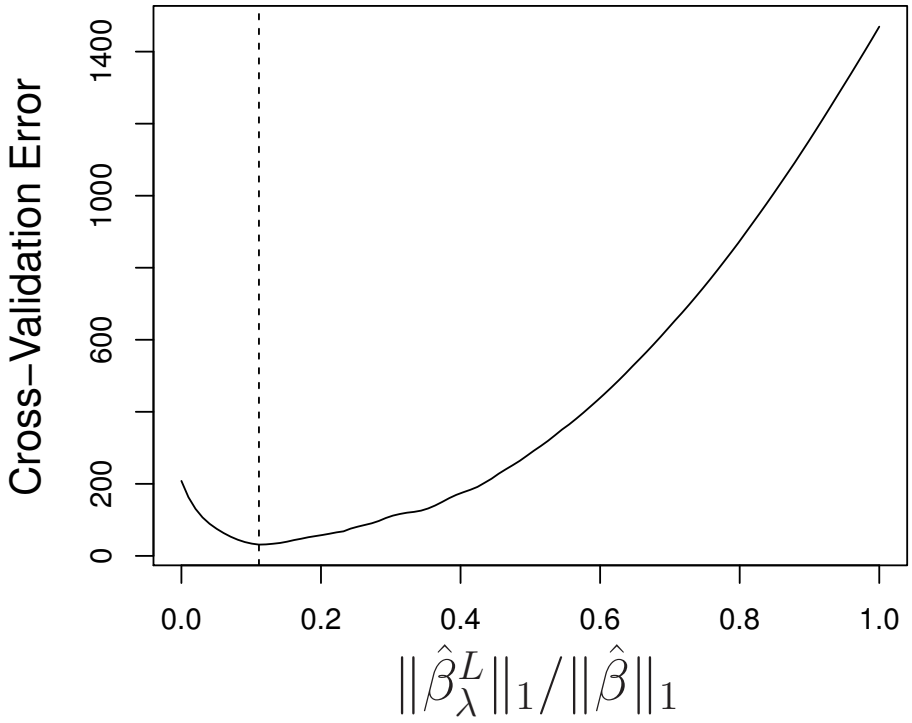
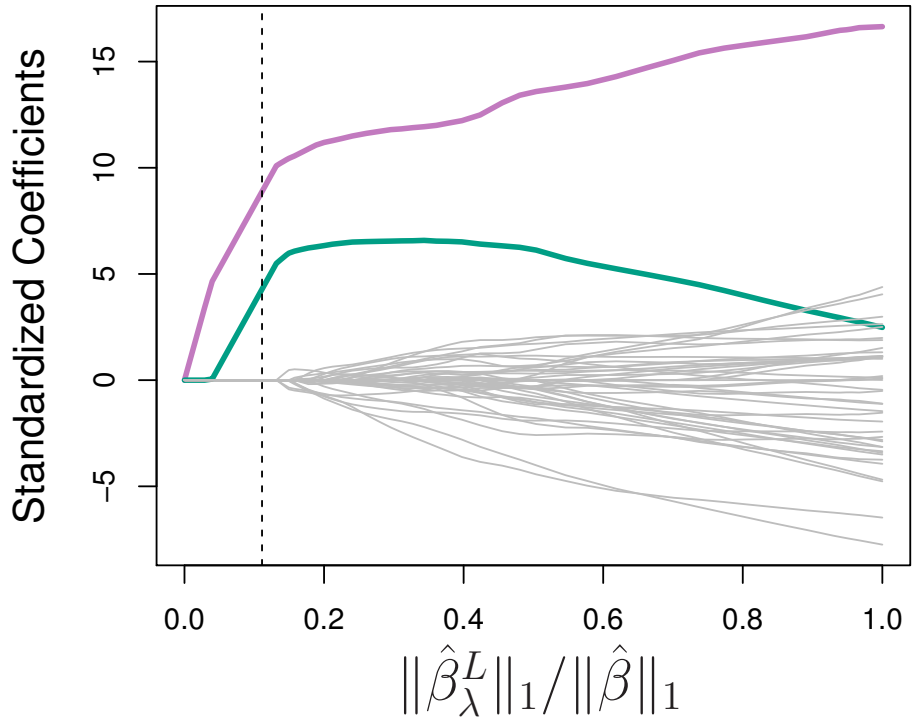
Simulated data set: 10-fold cross-validation MSE for LASSO.
Vertical dash lines indicate the lasso fit for which the cross-validation error is the smallest.





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RIDGE Regression or LASSO Regression?

- LASSO regression produces more interpretable models than ridge regression.
- Both regressions have similar behavior with respect to the squared bias.
- Ridge regression produces smaller MSE when all the regressors are related with the response variable.
- LASSO regression produces smaller MSE when only a subset of the regressors are related with the response variable.
- The number of predictors that is related to the response is never known a priori for real data sets.
- Cross-validation can be used in order to determine which approach is better on a data set.

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Bayesian view of ridge and lasso

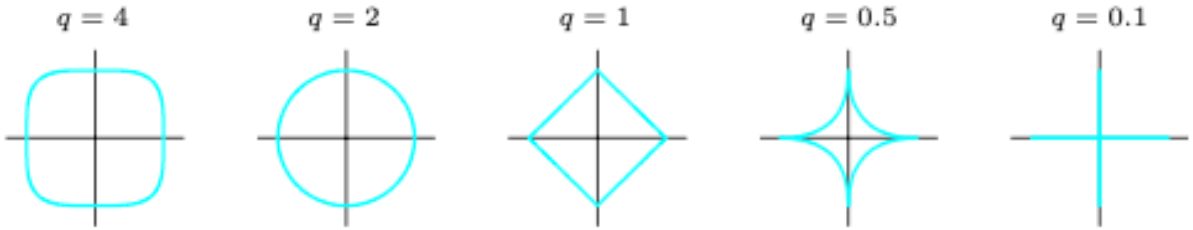
- We can generalize ridge regression and the lasso, viewed them as Bayes estimates:

$$\tilde{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta|_j^q$$

for $q \geq 0$.

- The contours of constant value of $\sum_{j=1}^{p} |\beta|_j^q$ are the **equi-contours of prior distribution of the parameters**, thinking of $|\beta|_j^q$ as the log-prior density for β_j.
- $q = 0$ corresponds to best subset selection, as the penalty simply counts the number of nonzero parameters;
- $q \leq 1$ the prior is not uniform in direction but concentrates more mass in the coordinate directions;
- $q = 1$ corresponds to the LASSO, i.e., the smallest q such that the constraint region is convex; non-convex constraint regions make the optimization problem more difficult.
- $q = 2$ Ridge regression.

$p = 2$



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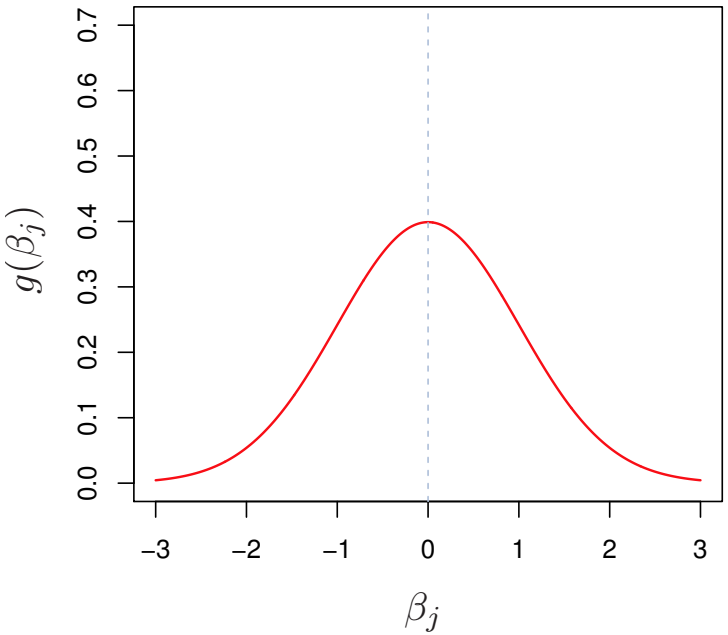
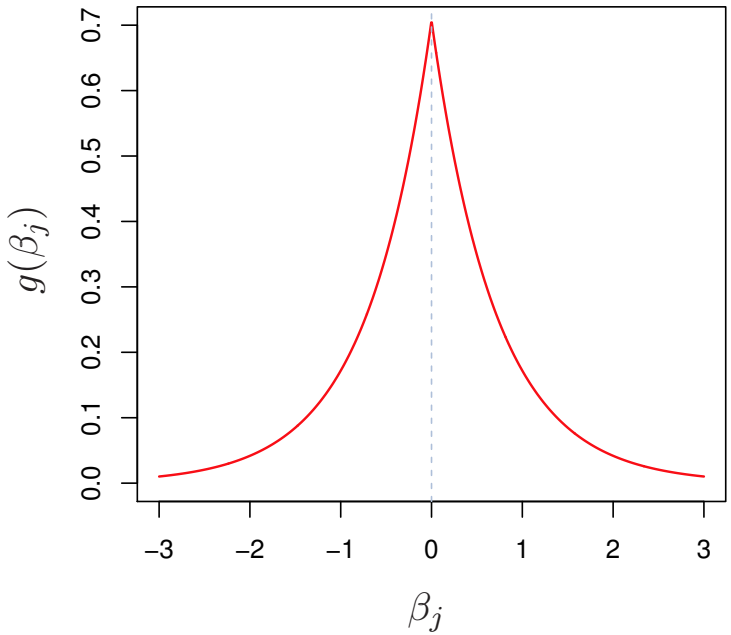
Bayesian Interpretation for Ridge and LASSO



Ridge is the posterior mean for β under a Gaussian prior



LASSO is the posterior mode for β under a double-exponential prior

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Toward a combination of ridge and lasso

- The lasso, ridge regression and best subset selection are Bayes estimates with different priors. Note, however, that they are derived as posterior modes, that is, maximizers of the posterior. It is more common to use the mean of the posterior as the Bayes estimate. Ridge regression is also the posterior mean, but the lasso and best subset selection are not.
- Looking again at the Bayesian criterion, we might try using other values of q besides $0, 1, \text{or } 2$. Although one might consider estimating q from the data, our experience is that it is not worth the effort for the extra variance incurred. Values of $q \in [1,2]$ suggest a compromise between the lasso and ridge regression. Although this is the case, with $q > 1$, $|\beta|_j^q$ is differentiable at 0, and so does not share the ability of lasso ($q = 1$) for setting coefficients exactly to zero.

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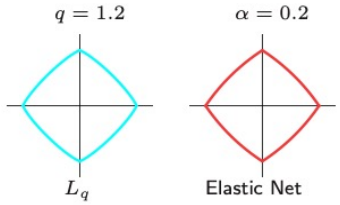
Elastic-Net Regression (Zou and Hastie, 2005)



- Elastic-Net penalty

$$\lambda \sum_{j=1}^{p} \left(\alpha \beta_j^2 + (1-\alpha)|\beta_j|\right)$$

- which is a different compromise between Ridge and Lasso.
- The elastic-net selects variables like the Lasso and shrinks together the coefficients of correlated predictors like Ridge.
- The first term encourages a sparse solution in the coefficients of these averaged features, while the second term highly correlated features to be averaged.
- It also has considerable computational advantages over the L_q penalties.
- The elastic-net can be used with any linear model, for regression as well as for classification.

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Elastic-net Optimization problem

- Elastic-net method solves the following regularization problem:

$$\min_{\beta_0,\beta}\left\{\frac{1}{2n}\sum_{i=1}^{n}(y_i-\beta_0-x_i'\beta)^2+\lambda P_\alpha(\beta)\right\}$$

For $0 \le \alpha \le 1$ and $\lambda \ge 0$, where

$$P_\alpha(\beta)=\frac{1-\alpha}{2}\|\beta\|_2^2+\alpha\|\beta\|_1=\sum_{j=1}^{p}\left(\frac{1-\alpha}{2}\beta_j^2+\alpha|\beta_j|\right)$$

- Elastic-net is equivalent to lasso when $\alpha = 1$. As α shrinks toward 0, elastic net approaches ridge regression. For other values of α, the penalty $P_\alpha(\beta)$ interpolates between the ℓ_1 norm of β and the ℓ_2 norm of β.

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Which one to be used

- The bias-variance tradeoff is a tradeoff between a complicated and simple model, in which an intermediate complexity is likely best.
- Lasso, Ridge Regression, and Elastic Net are modifications of ordinary least squares linear regression, which use additional penalty terms in the cost function to keep coefficient values small and simplify the model.
- Lasso is useful for feature selection, when our dataset has features with poor predictive power.
- Ridge regression is useful for the grouping effect, in which colinear features can be selected together.
- Elastic Net combines Lasso and ridge regression, potentially leading to a model that is both simple and predictive.

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«glmnet» for Elastic-net Regression using R

We use the same data as used before.

We run the Ridge regression on the learning set

```
> ridge.mod=glmnet(X[train,],Y[train],alpha=0)
```

We proceed with 10-fold cross-validation

```
> set.seed(1) #for reproducibility
> cvee=cv.glmnet(X[train,],Y[train],alpha=0)
```

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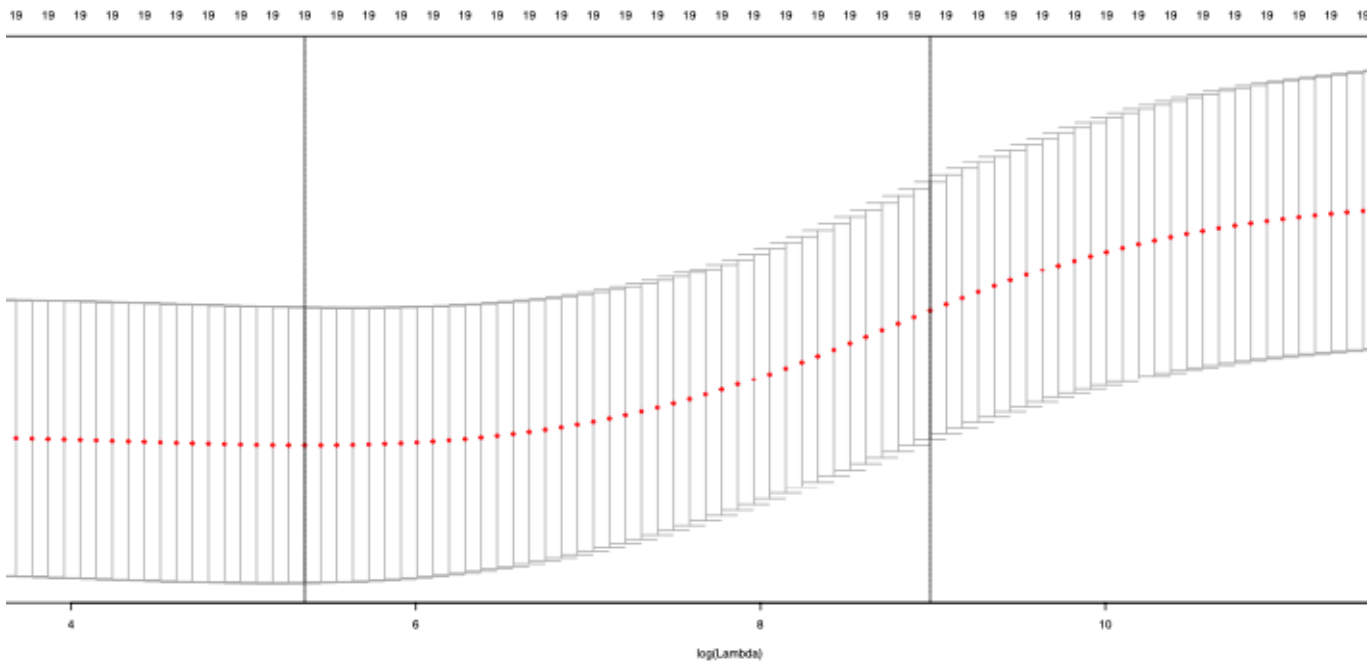

0-SE	1-SE
211.7416	7971.9347

Cross-validation with 0-SE and 1-SE selection

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```
> out=glmnet(X,Y,alpha=0)
> predict(out,type="coefficients",s=cvee$lambda.min)[1:20,]
(Intercept)        AtBat         Hits        HmRun         Runs          RBI
 9.88487157   0.03143991   1.00882875   0.13927624   1.11320781   0.87318990
      Walks        Years       CAtBat        CHits       CHmRun        CRuns
 1.80410229   0.13074381   0.01113978   0.06489843   0.45158546   0.12900049
       CRBI       CWalks      LeagueN    DivisionW      PutOuts      Assists
 0.13737712   0.02908572  27.18227535 -91.63411299   0.19149252   0.04254536
     Errors   NewLeagueN
-1.81244470   7.21208390
```

Ridge regression coefficients (λ selected with O-SE rule)

Hitter Data set: Salary as target

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LASSO Regression in R

```
> lasso.mod=glmnet(X[train,],Y[train],alpha=1)
```

```
> summary(lasso.mod)
          Length Class     Mode
a0          89   -none-    numeric
beta      1691   dgCMatrix S4
df          89   -none-    numeric
dim          2   -none-    numeric
lambda      89   -none-    numeric
dev.ratio   89   -none-    numeric
nulldev      1   -none-    numeric
npasses      1   -none-    numeric
jerr         1   -none-    numeric
offset       1   -none-    logical
call         4   -none-    call
nobs         1   -none-    numeric
```

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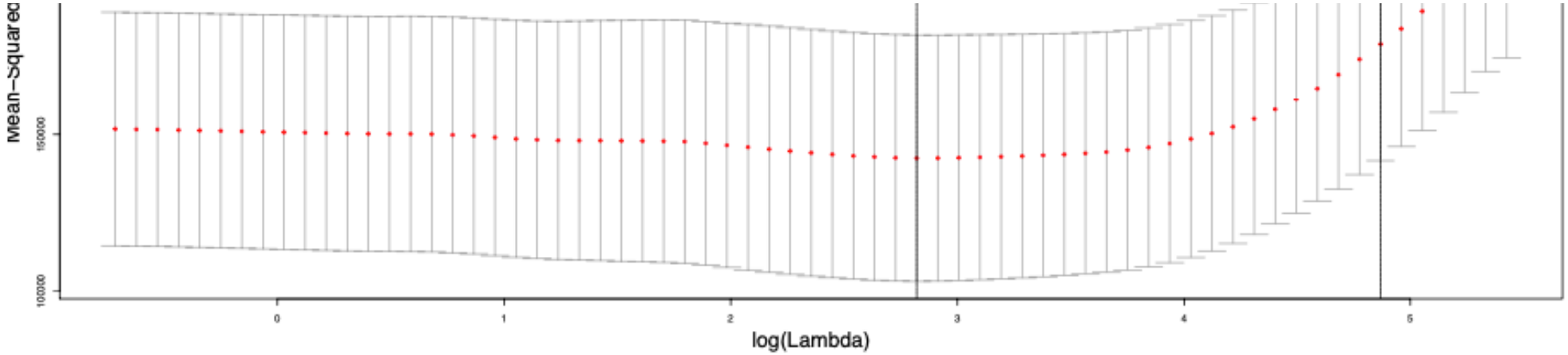
10-fold cross-validation with 0-SE and 1-SE selection

```
> set.seed(1) #for reproducibility
> cveeL=cv.glmnet(X[train,],Y[train],alpha=1)
```



0-SE 1-SE

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```
(Intercept)       AtBat        Hits       HmRun        Runs         RBI
 19.5223994   0.0000000   1.8701714   0.0000000   0.0000000   0.0000000
      Walks       Years      CAtBat       CHits      CHmRun       CRuns
  2.2187934   0.0000000   0.0000000   0.0000000   0.0000000   0.2072852
       CRBI      CWalks     LeagueN   DivisionW     PutOuts     Assists
  0.4127984   0.0000000   1.7591971 -103.5051402   0.2206884   0.0000000
     Errors   NewLeagueN
  0.0000000   0.0000000
```

LASSO regression coefficients (λ selected with O-SE rule)

```
> out=glmnet(X,Y,alpha=1)
> predict(out,type="coefficients",s=cveeL$lambda.min)[1:20,]
```

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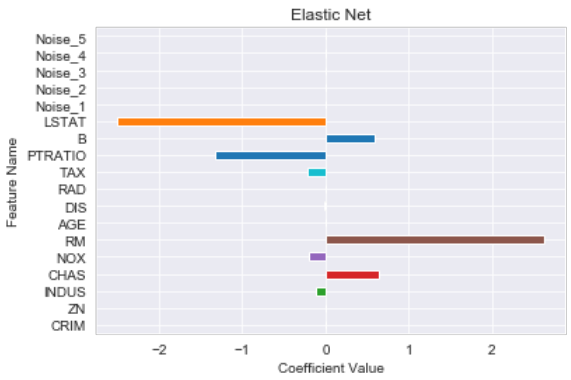
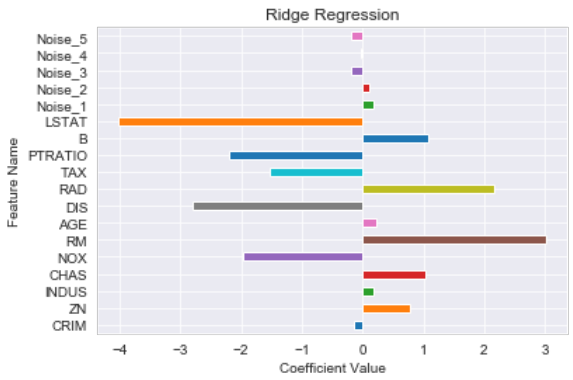
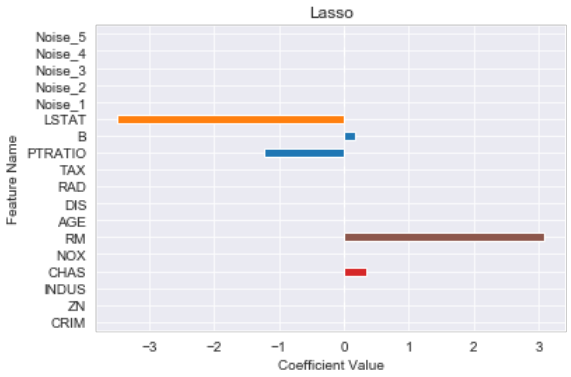


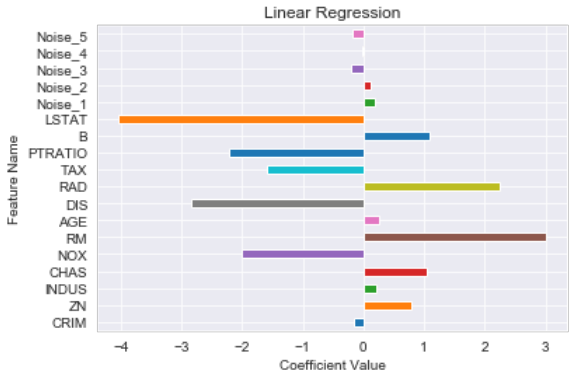
Test MSE = 24.63



Test MSE = 23.87



Test MSE = 24.91



Test MSE = 23.87

Boston Housing Data

UCI Machine Learning Repository

[Scikit-learn: Machine Learning in Python]

- Using the Boston Housing Dataset available in ***sklearn***, we will examine the results of all 4 of our algorithms.
- On top of this data, I scaled the data and created 5 additional 'features' of random noise to test each algorithm's ability to filter out irrelevant information.
- I will not do any parameter tuning; I will just implement these algorithms out of the box.
- You can see default parameters in ***sklearn***'s documentation.

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Dimension Reduction Methods

- Methods seen before work with original predictors.
- Now we see a class of methods that transform the predictors and then fit a least squares model using the transformed variables.
- We will refer to these to
  - **Principal Component Regression**
  - **Partial Least Squares regression**
- These are not variable selection as original predictors are all considered.

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Principal Component Regression

- **Principal components analysis (PCA)** is a popular approach for deriving a low-dimensional set of features from a large set of variables.
- The idea underlying the Principal Components Regression (PCR) is computing the first M principal components, $Z_1, \ldots, Z_M$, and then using these components as the predictors in a linear regression model that is fit using least squares.
- This can be done because often ***a small number of components explains a large amount of variability.***

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Properties of Principal Component Regression

- ✓ The basic assumption is that **the directions in which $X_1, \dots, X_p$ show the most variation are the directions that are associated with the response variable.**
- ✓ The number of principal components to be chosen in performing a PCR is typically chosen by **cross-validation**.
- ***Drawback:*** there is no guarantee that the directions that best explain the predictors will also be the best directions to use for predicting the response.

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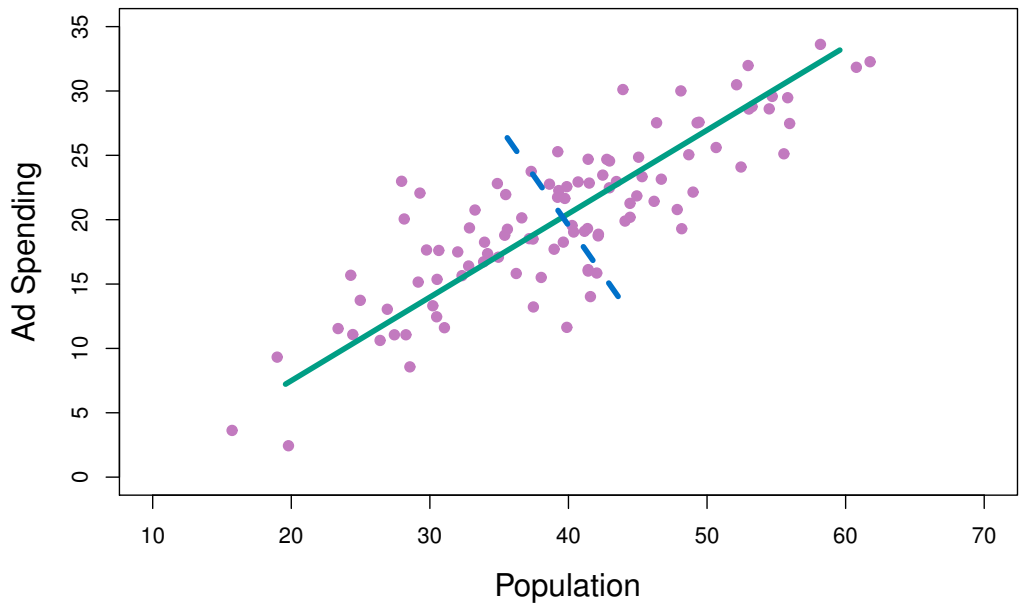
PCA on Population size and Ad Spending for 100 cities



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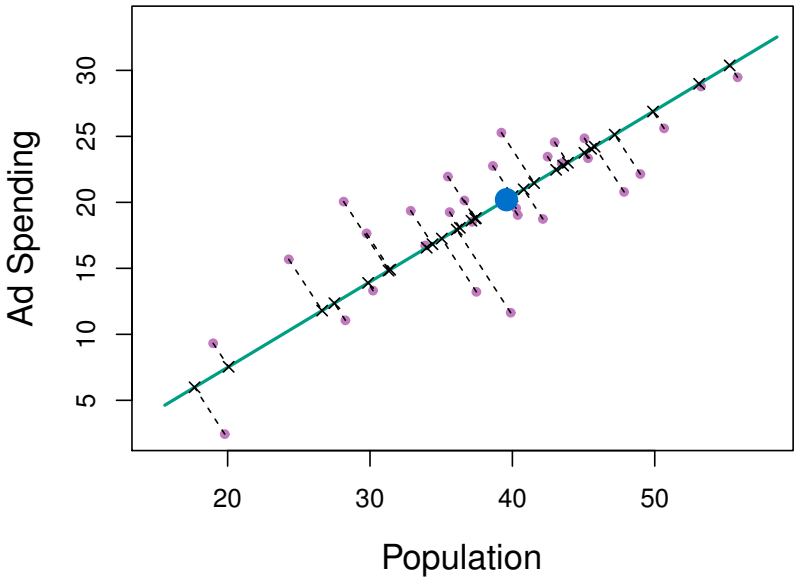
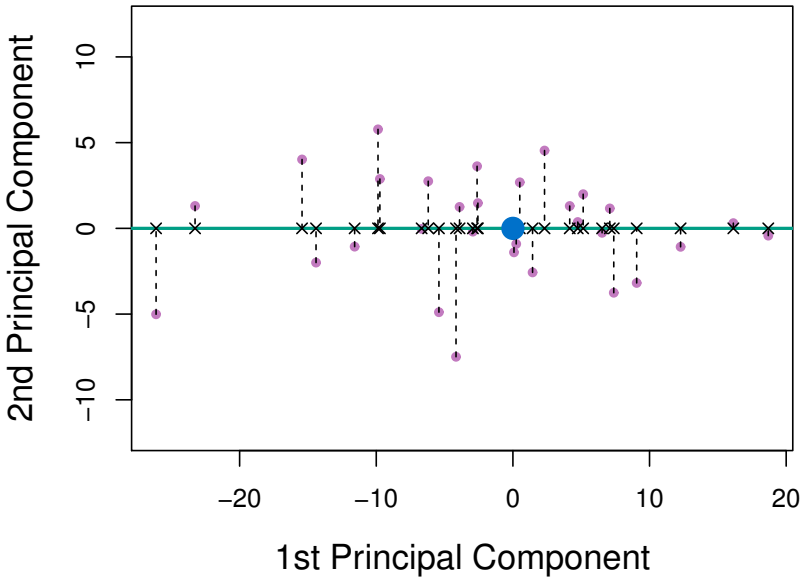
Most of the variance captured by the first PC





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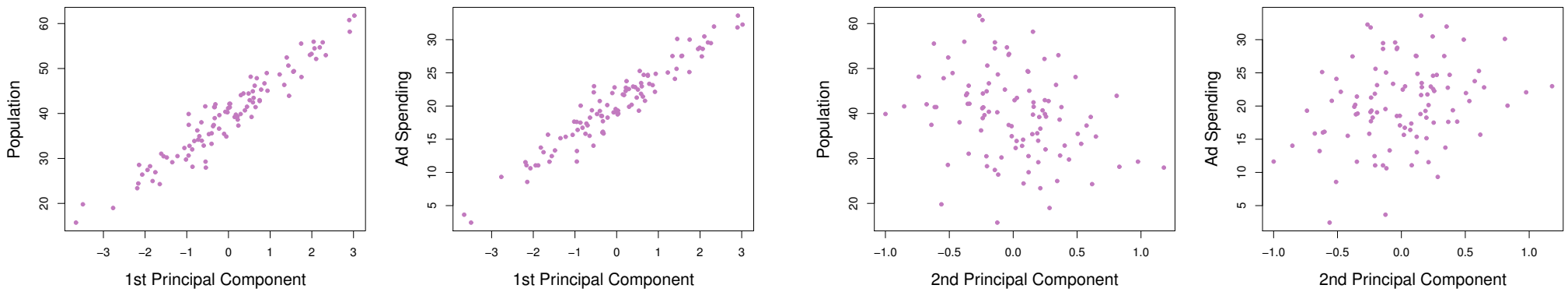
Relationships of the PCs with the original features



✓ Original predictors need to be standardized

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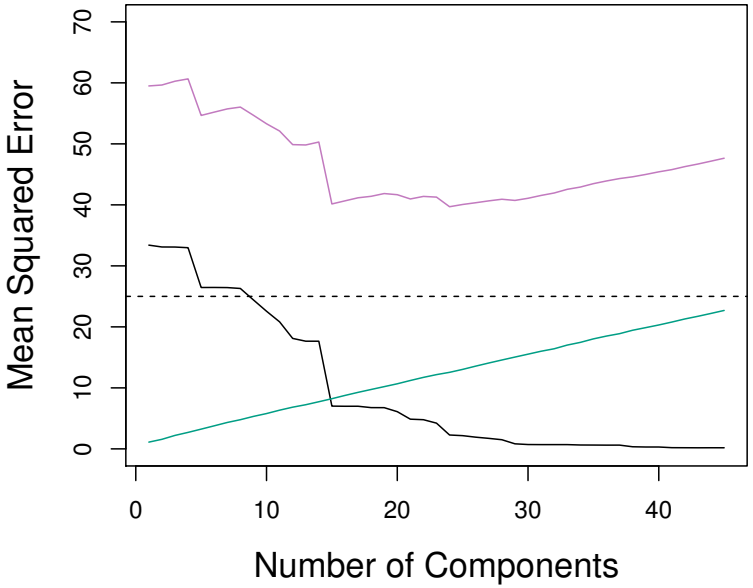
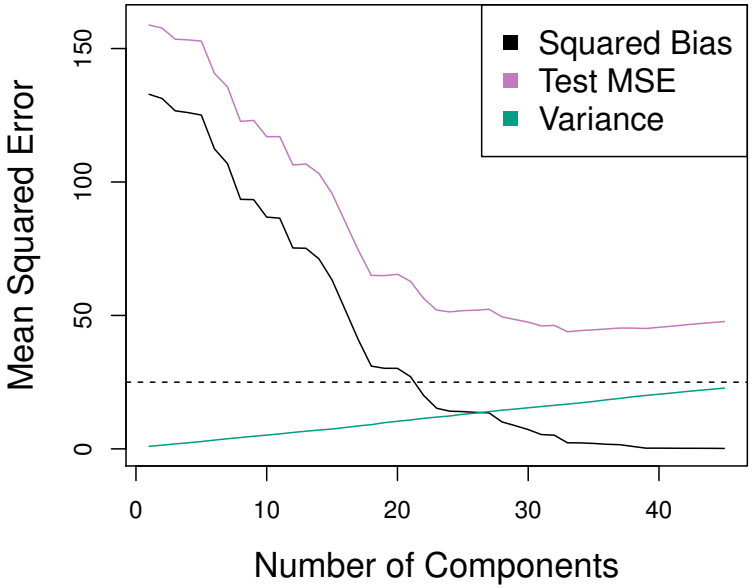
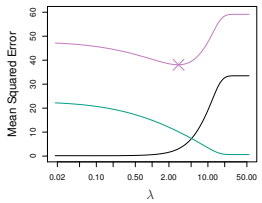
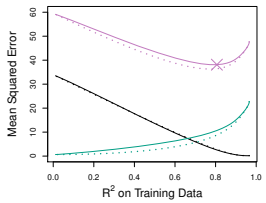
PCR Performance on Simulated Data Sets previously used for LASSO - n = 50, p = 45 (left), p = 2 (right) -

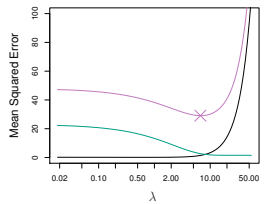


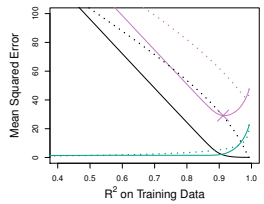






PCR goes worse as more PCs are necessary to capture the overall variation





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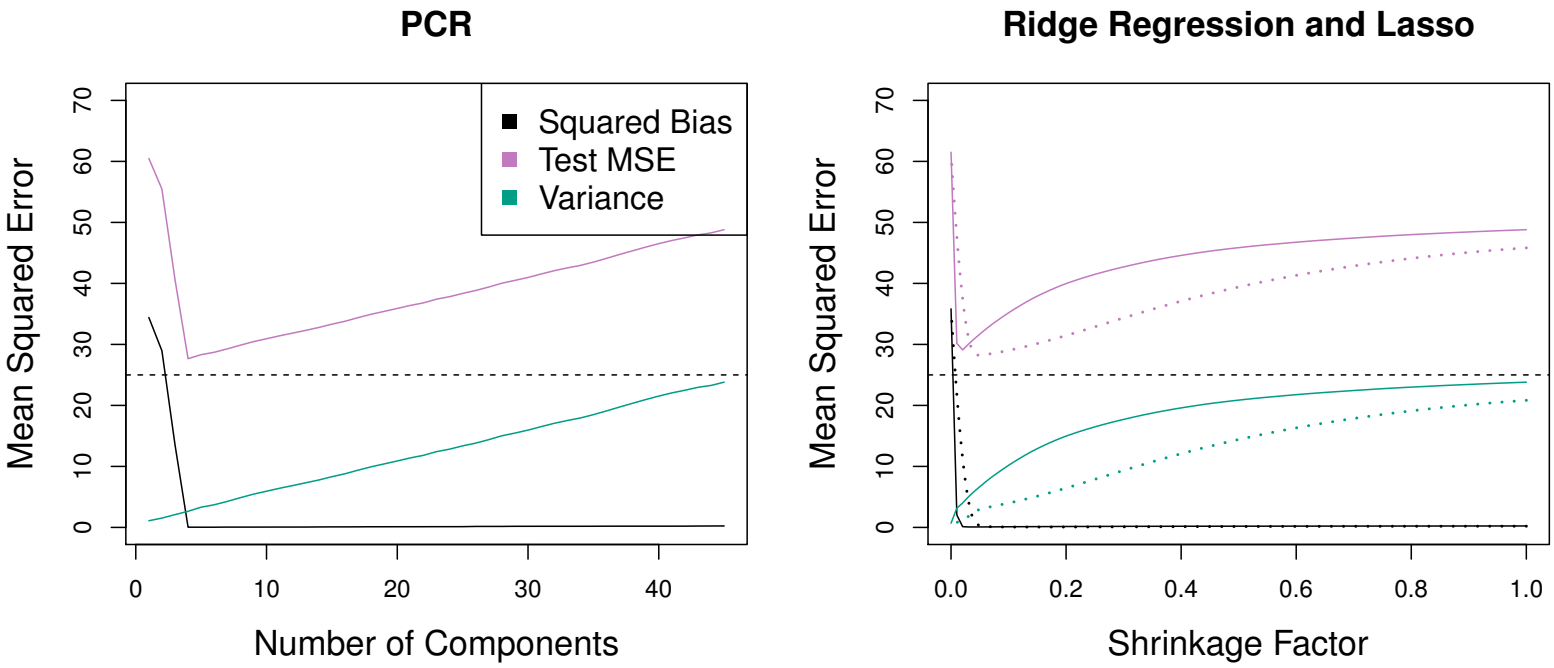
PCR, Ridge, LASSO on Simulated Data Set



PCR goes well as few PCs (five) capture most of the variation
PCR and Ridge (dots) slightly outperform LASSO (solid)

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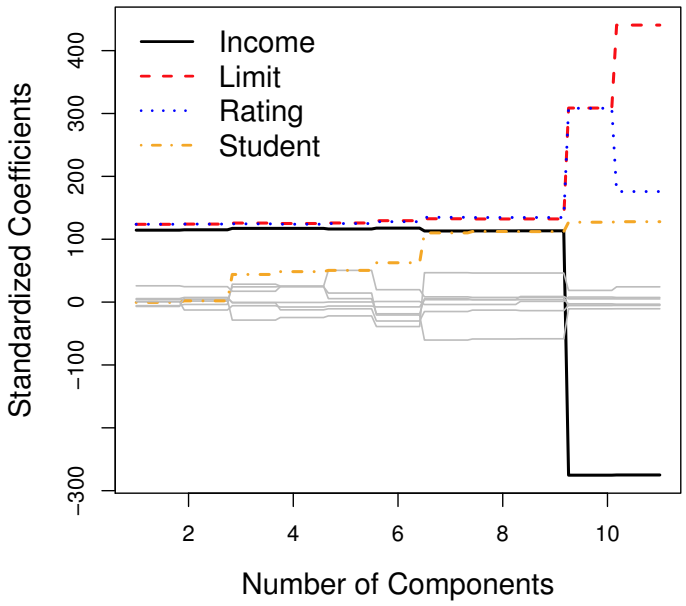
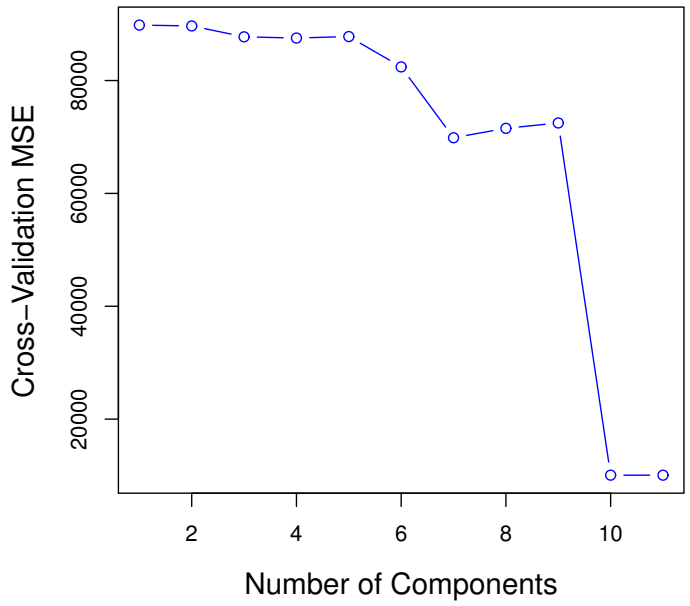
PCR standardized coefficient estimates and 10-fold Cross Validation MSE for Credit data set





The lowest Cross Validation occurs for 10 PCs, almost near to 11 original predictors so that no dimensionality reduction is provided

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Partial Least Squares Regression

In PCR the response variable is not used to help determining the principal component directions.

Partial Least Squares (PLS) regression is a supervised alternative to PCR. Like PCR, PLS is a dimension reduction method, which first identifies a new set of features that are linear combinations of the original features, and then fits a linear model via least squares.

Unlike PCR, PLS makes use of the response variable in order to identify new features that not only approximate the old features well, but also that are related to the response. The PLS approach attempts to find directions that help explain both the response and the predictors.

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Partial Least Squares Regression: the Algorithm

- Initialization: standardize each of the p predictors in order to have mean zero and unit variance. Set $\hat{\mathbf{y}}^{(0)} = \bar{y}\mathbf{1}$ and $\mathbf{x}_j^{(0)} = \mathbf{x}_j$, $\forall j = 1, \dots, p$.
- For $\mathrm{m} = 1, \dots, p$
  1. $\mathbf{z}_m = \sum_{j=1}^{p} \hat{\phi}_{m_j} \mathbf{x}_j^{(m-1)}$, with $\hat{\phi}_{m_j} = \left\langle \mathbf{x}_j^{(m-1)}, \mathbf{y} \right\rangle$
  2. $\hat{\theta}_m = \mathbf{z}_m' \mathbf{y} \,/ \mathbf{z}_m' \mathbf{z}_m$
  3. $\hat{\mathbf{y}}^{(m)} = \hat{\mathbf{y}}^{(m-1)} + \hat{\theta}_m \mathbf{z}_m$
  4. Orthogonalize each $\mathbf{x}_j^{(m-1)}$ with respect to $\mathbf{z}_m$:

  $\mathbf{x}_j^{(m)} = \mathbf{x}_j^{(m-1)} - \left[\mathbf{z}_m' \mathbf{x}_j^{(m-1)} / \mathbf{z}_m' \mathbf{z}_m\right] \mathbf{z}_m \,, j = 1, \dots, p$
- Output the sequence of fitted vectors $\hat{\mathbf{y}}^{(m)}$.
- Since the $\mathbf{z}_l$ for $\mathrm{l} = 1, \dots, m$ are linear in the original $\mathbf{x}_j$, so is $\hat{\mathbf{y}}^{(m)} = \mathbf{X}\hat{\beta}^{PLS}(m)$.
- These linear coefficients can be recovered from the sequence of PLS transformations.

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Properties of PLS Regression

- Since it uses the response variable $\mathbf{y}$ to construct its directions, Partial Least Squares solution is a nonlinear function of $\mathbf{y}$.
- Partial Least-Squares seeks directions that have high variance and have high correlation with the response $\mathbf{y}$, in contrast to PCR.

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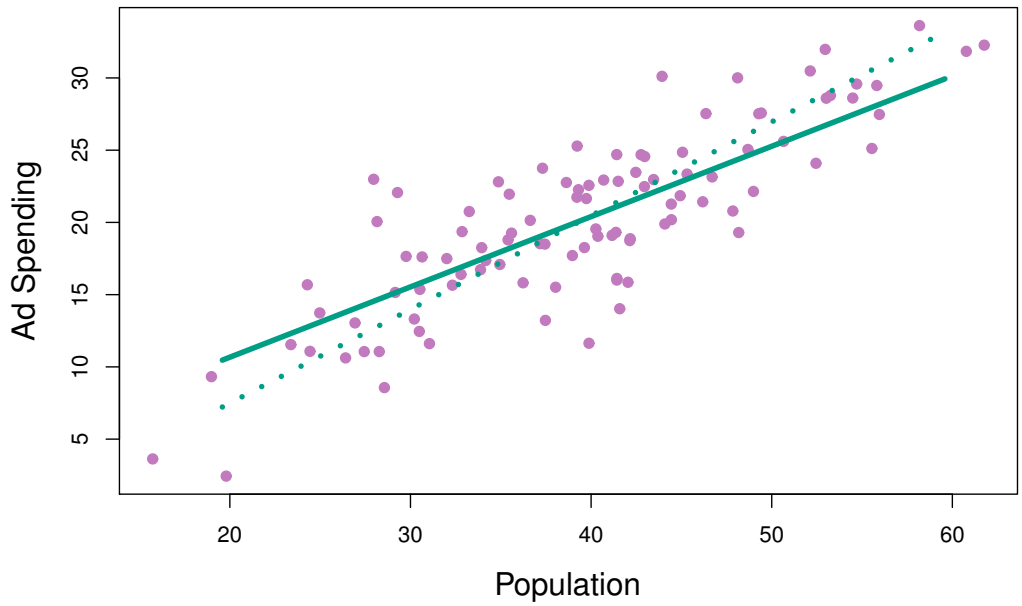
PLS first direction for advertising data



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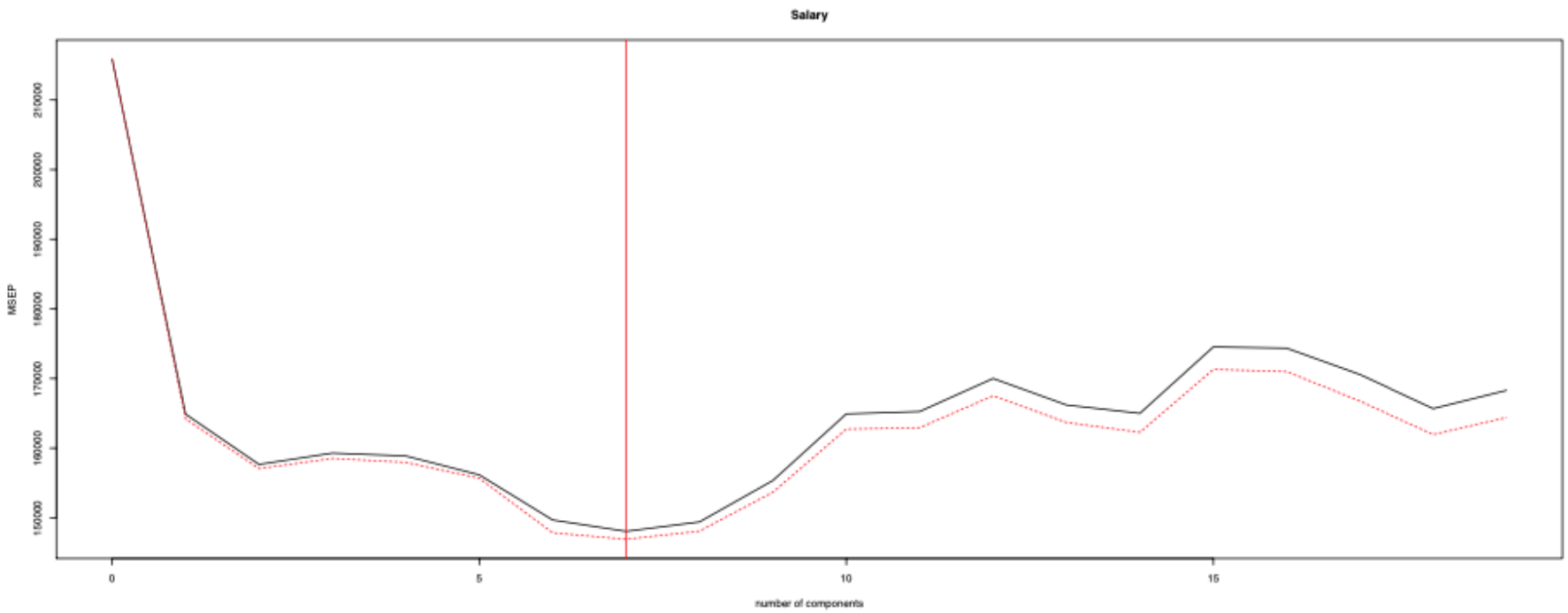

PCR in R

Run PCR on Hitters (training) data set, by choosing the number of components through cross-validation

```
> #require(pls)
> set.seed(1) #for reproducibility
> pcreg=pcr(Salary~., data=Hitters, subset=train, + scale=TRUE, validation="CV")
```

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PCR in R

- Test set MSE:

```
> x=model.matrix(Salary~.,Hitters)[,-1]
> PCRpred = predict (pcreg, x[test,], ncomp = 7) >
MSEtest=mean((PCRpred-
Hitters$Salary[test])^2) > MSEtest
```

[1] 96556.22

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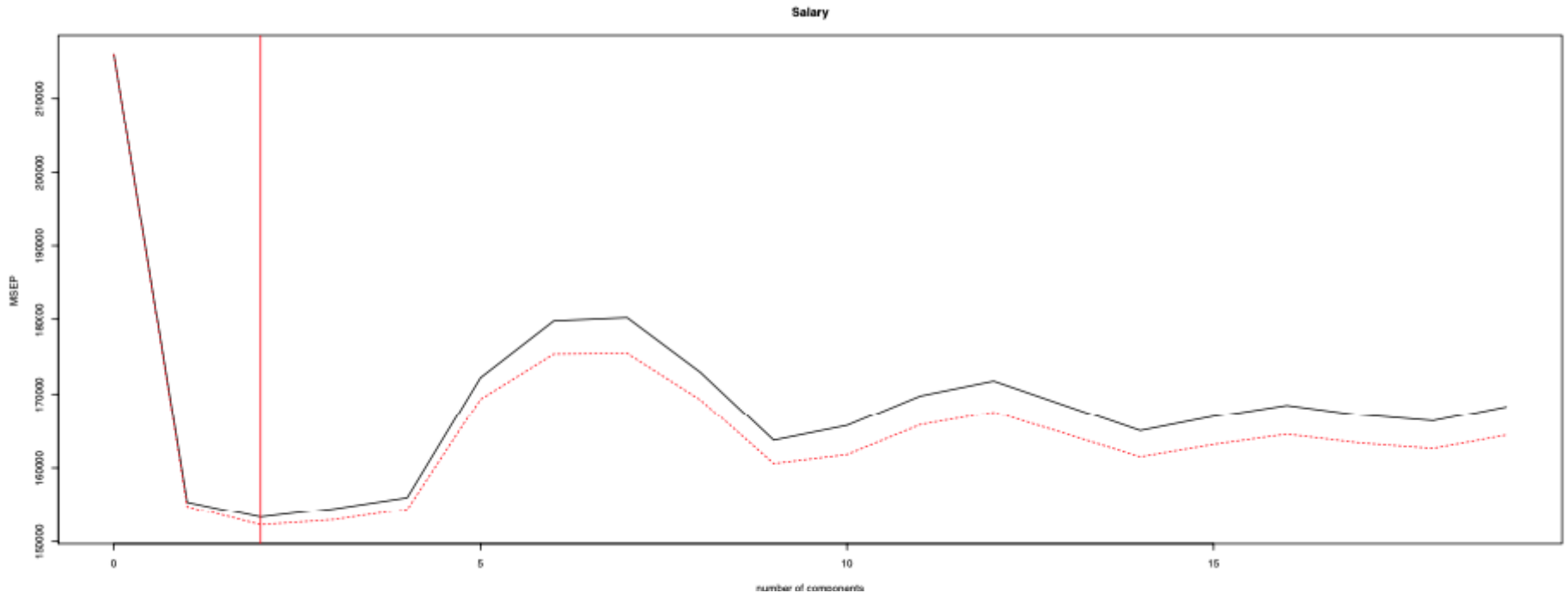

PLS in R

Run PLS regression on Hitters (training) data set, by choosing the number of components through cross-validation

```
> #require(pls)
> set.seed(1) #for reproducibility
> plsreg=plsr(Salary~., data=Hitters, subset=train, + scale=TRUE, validation="CV")
```

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PLS in R

- Test set MSE:

```
> PLSpred = predict (plsreg, x[test,], ncomp = 2) >
MSEtest=mean((PLSpred-Hitters$Salary[test])^2)
> MSEtest

[1] 101417.5
```

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What about High-Dimensional Data?

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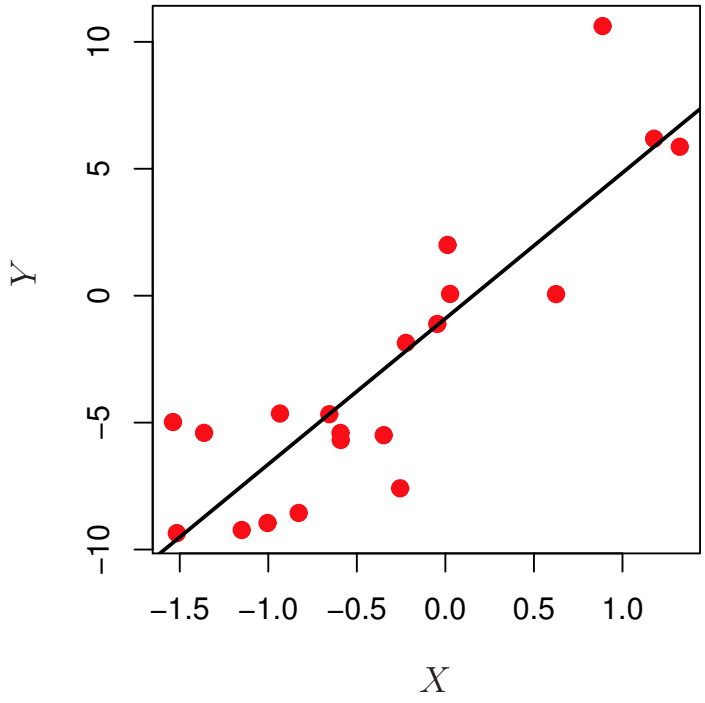

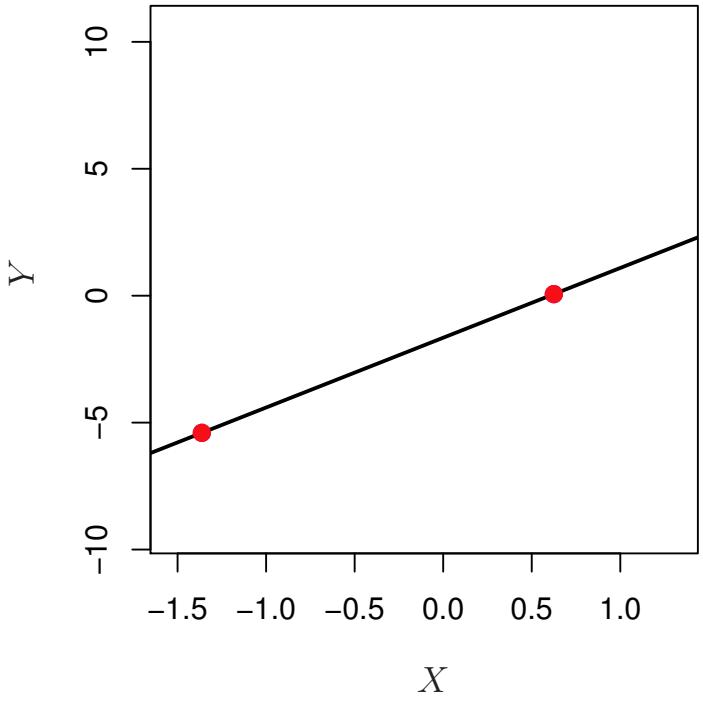

About the flexibility

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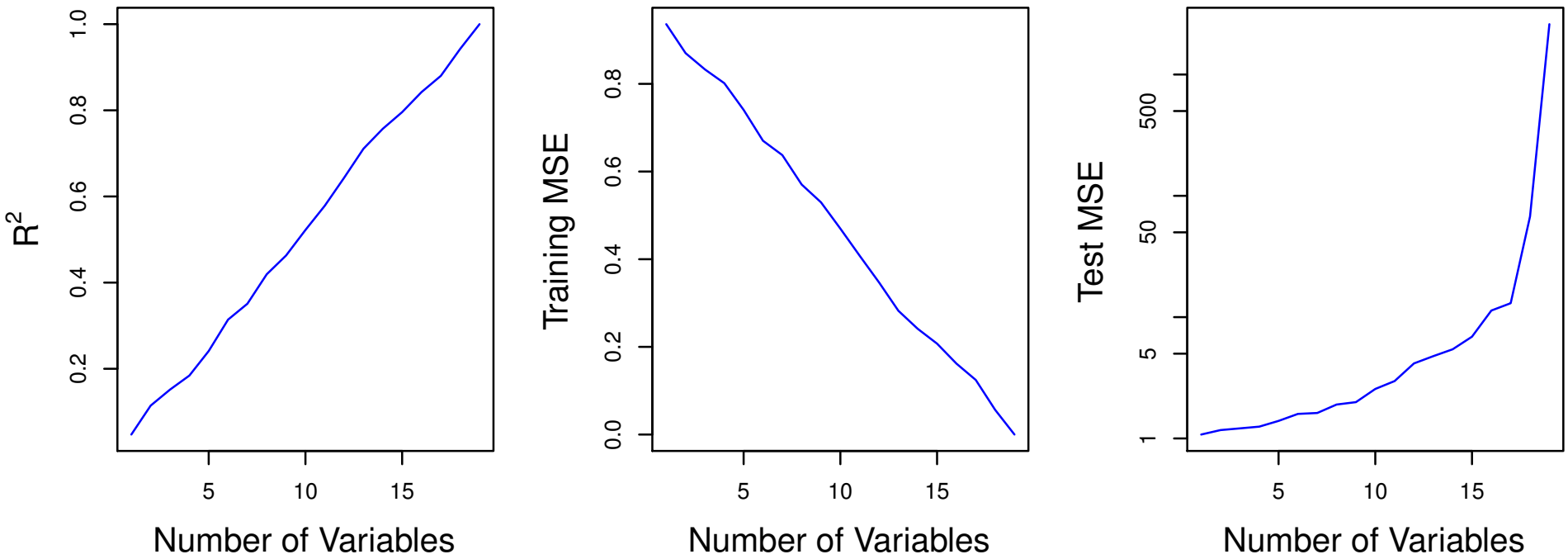

Flexibility yields to overfitting

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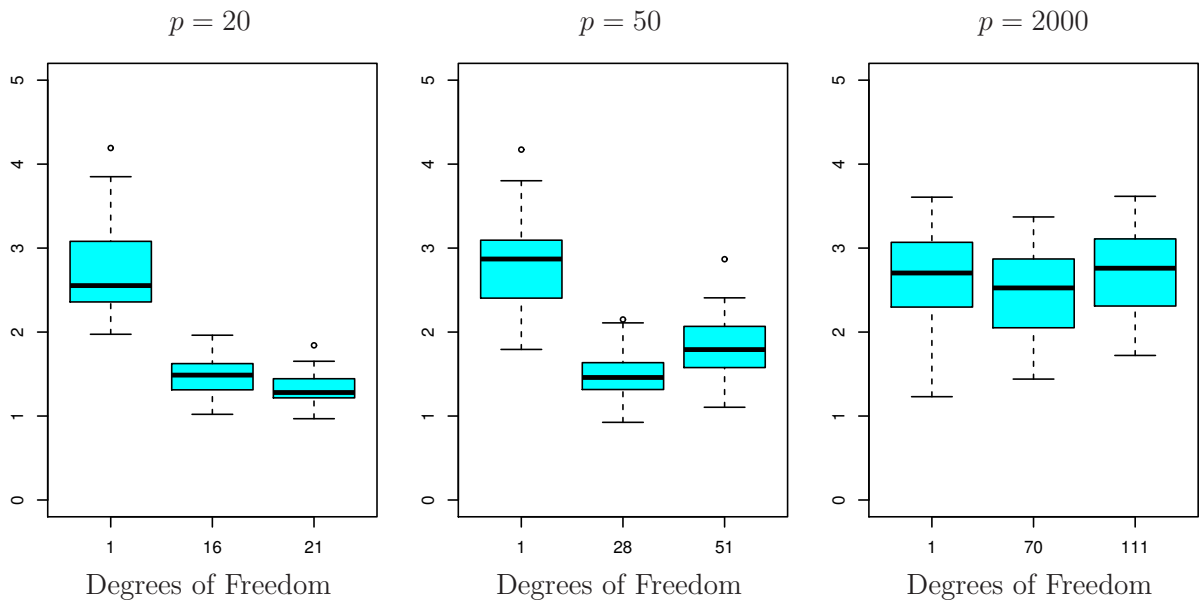
Regression in high dimension
LASSO performance in terms of test MSE

- The lasso was performed with $n = 100$ observations and three values of p, the number of features. Of the p features, 20 were associated with the response.
- The boxplots show the test MSEs that result using three different values of the tuning parameter λ . For ease of interpretation, rather than reporting λ, the degrees of freedom are reported; for the lasso this turns out to be simply the number of estimated non-zero coefficients.
- When $p = 20$, the lowest test MSE was obtained with the smallest amount of regularization.
- When $p = 50$, the lowest test MSE was achieved when there is a substantial amount of regularization.
- When $p = 2{,}000$ the lasso performed poorly regardless of the amount of regularization, due to the fact that only 20 of the 2,000 features truly are associated with the outcome.



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Three important points

Regularization or shrinkage plays a key role in high-dimensional problems

Appropriate tuning parameter selection is crucial for good predictive performance

The test error tends to increase as the dimensionality of the problem (i.e., the number of features) increases, unless the additional features are truly associated with the response.

Adding additional signal features that are truly associated with the response will improve the fitted model (yielding to reduce the test set error).

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